

Q12

```
#include <stdio.h>
int main()
{
    char entry;
    printf("enter a character");
    scanf("%c", &entry);
    printf("ascii value of '%c' = '%d'", entry, entry);
}
```

```
PS E:\intro to prog assignments sem1> ${env:
nsions\ms-vscode.cpptools-1.34.4-win32-x64\de
soft-MIEngine-In-epw4mxx3.rsv' '--stdout=Micro
ngine-Error-qxv5fqrt.a1l' '--pid=Microsoft-MI
gdb.exe' '--interpreter=mi'
enter a charactera
ascii value of 'a' = '97'
PS E:\intro to prog assignments sem1> █
```

Q13

```
#include <stdio.h>
int main()
{
    int number;
    printf("enter a number");
    scanf("%d", &number);
    if(number%2==0)
    {
        printf("entered number is even");
    }
    else
    {
        printf("entered number is odd");
    }
}
```



```
gdb.exe' '--interpreter=mi'  
enter a number 13  
entered number is odd  
PS E:\intro to prog assignments sem1>
```

```
gdb.exe' '--interpreter=mi'  
enter a number 14  
entered number is even  
PS E:\intro to prog assignments sem1>
```

Q14

```
#include <stdio.h>  
int main()  
{  
    char enter;  
    printf("enter a character\n");  
    scanf("%c", &enter);  
    if (enter >= 'A' && enter <= 'Z')  
    {  
        printf("Uppercase Letter");  
    }  
    else if(enter >='a' && enter <='z')  
    {  
        printf("Lowercase Letter");  
    }  
    else if(enter >='0' && enter <='9')  
    {  
        printf("digit");  
    }  
    else  
    {  
        printf("you have entered a special symbol");  
    }  
    return 0;  
}
```



```
gdb.exe' --interpreter
enter a character
S
Uppercase Letter
```

```
enter a character
i
Lowercase Letter
```

```
gdb.exe' --interpreter=mi
enter a character
&
you have entered a special symbol
```

Q15

```
#include <stdio.h>

int main() {
    int sub1, sub2, sub3, sub4, sub5, total;
    printf("enter your marks in subject-1,2,3,4,5(out of 100 in each)");
    scanf("%d %d %d %d %d", &sub1, &sub2, &sub3, &sub4, &sub5);
    total = sub1+sub2+sub3+sub4+sub5;
    if(total >=450 && total <= 500)
    {
        printf("first division");
    }
    else if(total>=330 && total <= 449)
    {
        printf("second division ");
    }
    else if(total >=175 && total <= 329)
    {
        printf("third division");
    }
    else if(total >= 0 && total <= 174)
    {
        printf("failed");
    }
}
```



```
gdb.exe' '--interpreter=mi'
enter your marks in subject-1,2,3,4,5(out of 100 in each)50 40 60 70 70
third division
PS E:\intro to prog assignments sem1> |
```

Q16

```
#include <stdio.h>
int main()
{
    int num1,num2,num3;
    printf("enter three distinct numbers:");
    scanf("%d %d %d", &num1,&num2,&num3);
    if (num1>num2);
    {
        if (num1>num3)
        {
            printf("%d is the greatest number", num1);
        }
    }
    if (num2>num1);
    {
        if (num2>num3)
        {
            printf("%d is the greatest number", num2);
        }
    }
    if (num3>num1);
    {
        if (num3>num2)
        {
            printf("%d is the greatest number", num3);
        }
    }
}
```



```
gdb.exe --interpreter=mi
enter three distinct numbers: 10 11 14
14 is the greatest number
PS E:\intro to prog assignments sem1>
```

Q17

```
#include <stdio.h>
int main()
{
    int a,b,c,max;
    printf("enter three distinct numbers");
    scanf("%d %d %d", &a,&b,&c);
    if(a>b && a>c)
    {
        max = a;
    }
    else if(b>a && b>c)
    {
        max = b;
    }
    else if(c>a && c>b)
    {
        max = c;
    }
    printf("%d is the greatest out of the three given numbers", max);
}
```

```
gdb.exe --interpreter=mi
enter three distinct numbers 13 12 15
15 is the greatest out of the three given numbers
PS E:\intro to prog assignments sem1>
```

Q18

```
#include <stdio.h>
int main()
```



```
{
    int a,b,c,max;
    printf("enter three distinct numbers");
    scanf("%d %d %d",&a,&b,&c);
    a>b && a>c?printf("%d is greatest",a):b>c && b>a?printf("%d is
greatest",b):c>a && c>b?printf("%d is greatest",c):printf("invalid numbers
entered");
}
```

```
gdb.exe' '--interpreter=mi'
enter three distinct numbers 12 10 7
12 is greatest
PS E:\intro to prog assignments sem1>
```

Q19

```
#include <stdio.h>
int main()
{
    char enter;
    printf("enter a letter");
    scanf(" %c", &enter);
    enter >= 'a' && enter <= 'z'?printf("smaller case letter"):printf("not a
smaller case letter");
    return 0;
}
```

```
gdb.exe' '--interpreter=mi'
enter a letter b
smaller case letter
PS E:\intro to prog assignments sem1>
```

Q20

```
#include <stdio.h>
int main()
{
```



```

int operation;
printf("enter the number in front of operation to perform the
corresponding operation \n 1 addition\n 2 subtraction (first - second)\n 3
multiplication\n 4 division greatest integer(first/second)\n");
scanf("%d",&operation);
int a,b;
printf("enter two numbers");
scanf("%d %d",&a,&b);
switch(operation)
{
case 1 : printf("addition = %d",a+b);
        break;
case 2 : printf("subtraction = %d",a-b);
        break;
case 3 : printf("multiplication = %d",a*b);
        break;
case 4 : printf("division (greatest integer)",a/b);
        break;
default : printf("invalid operation number");
}
}

```

```

gdb.exe' '--interpreter=mi'
enter the number in front of operation to perform the corresponding operation
1 addition
2 subtraction (first - second)
3 multiplication
4 division greatest integer(first/second)
3
enter two numbers 12 11
multiplication = 132
PS E:\intro to prog assignments sem1>

```

Q21

```

#include <stdio.h>
int main()
{
    char operation;

```



```

printf("enter the symbol in front of operation to perform the
corresponding operation \n + addition\n - subtraction (first - second)\n *
multiplication\n / division greatest integer(first/second)\n");
scanf("%c",&operation);
int a,b;
printf("enter two numbers");
scanf("%d %d",&a,&b);
switch(operation)
{
case '+' : printf("addition = %d",a+b);
break;
case '-' : printf("subtraction = %d",a-b);
break;
case '*' : printf("multiplication = %d",a*b);
break;
case '/' : printf("division (greatest integer)",a/b);
break;
default : printf("invalid operation symbol");
}
}

```

```

enter the symbol in front of operation to perform the corresponding operation
+ addition
- subtraction (first - second)
* multiplication
/ division greatest integer(first/second)
+
enter two numbers 33 44
addition = 77
PS E:\intro to prog assignments sem1>

```

Q22

```

#include <stdio.h>

int main()
{
    long int num,fact=1,tpr;
    printf("enter a number");
    scanf("%ld",&num);
}

```



```

tpr = num;
while(num>0)
{
    fact= fact*num;
    num -=1;
}
printf("factorial of %ld is %ld",tpr,fact);
}

```

```

gdb.exe -i interpreter-4m
enter a number7
factorial of 7 is 5040
PS E:\intro to prog assignments sem1>

```

Q23

```

#include <stdio.h>
int main()
{
    int a,b,tpr,pow=1;
    printf("enter a number to be base ");
    scanf("%d",&a);
    printf("enter a number to be power ");
    scanf("%d",&b);
    tpr = b;
    while(b>0)
    {
        pow = pow*a;
        b -= 1;
    }
    printf("Base %d powered %d = %d",a,tpr,pow);
}

```



```
gub.exe --interpreter=ml
```

```
enter a number to be base 2
```

```
enter a number to be power 5
```

```
Base 2 powered 5 = 32
```

```
PS E:\intro to prog assignments sem1> █
```

